Problems 2 | Output | Debug Console | Terminal | Ports | Spell Checker 2 | DevDb | Console Ninja

pwsh

```
PS C:\Users\Laptop Solutions\Downloads\NAVTTC-A3> py student_management_oop.py
=== PYTHON OOP: STUDENT MANAGEMENT SYSTEM ===

A. FUNCTIONS AND FUNCTIONAL PROGRAMMING
Adding Adnan
Adding Sara
Student count (global): 2
Average: 84.33333333333333
Grade: A
Square of 6: 36
Marks out of 10: [8.5, 9.0, 7.8]
Students with average >= 50: ['Adnan', 'Sara']

B. FILE AND EXCEPTION HANDLING
File not found: missing_students_example.txt
File read attempt finished.
Read 2 record(s) from students.txt:
  101|Adnan|85,90,78
  102|Sara|45,65,55
File read attempt finished.
Invalid marks input 'abc': could not convert string to float: 'abc'
Validation attempt finished.
Invalid marks input '120': Marks must be between 0 and 100.
Validation attempt finished.
Validation attempt finished.
Valid mark: 75.0
Modes: r=read, w=write/replace, a=append, x=create only if absent.
```

```
C. CLASSES AND OBJECTS
Shared school name: National College National College
Adnan (Roll: 101), marks: [85.0, 90.0, 78.0], average: 84.33, grade: A
Public roll: 101 | protected marks: [85.0, 90.0, 78.0]
Private status via method: Active
Mangled attribute example: Active
Student __str__: Student(Adnan, roll=101) | __len__: 3

D. INHERITANCE AND MRO
Name: Bilal
Graduate research: Machine Learning
Student athlete sport: Cricket
StudentAthlete MRO: ['StudentAthlete', 'Student', 'Person', 'SportsMember', 'object']

E. CLASS RELATIONSHIPS
Aggregation: Computer Science has ['Adnan', 'Sara']
Association: Adnan is taught by Mr. Ahmed
Composition: Adnan report: 3 subjects, total 253.0
Nested ContactInfo: adnan@example.com, 0300-0000000
```

```
F. POLYMORPHISM AND OPERATOR OVERLOADING
Student: Adnan, roll 101
Student: Sara, roll 102
Teacher: Mr. Ahmed, teaches Python
Combined marks: Marks[80.0, 90.0, 70.0, 85.0] | count: 4

G. ABSTRACT, EMPTY AND DATA CLASSES
Adnan (Roll: 101), marks: [85.0, 90.0, 78.0], average: 84.33, grade: A
Empty class object: TemporaryRecord
Courses: Course(course_name='Python', course_code='CS101', credit_hours=3) Course(course_name='OOP', course_code='CS102', credit_h
ours=4)
Cannot create Record directly: display() is abstract.

H. INTEGRATION AND TESTING
Students: 2 | Teachers: 1
Valid data accepted; invalid marks and missing file handled.
Saved file: students.txt
Program completed successfully.
PS C:\Users\Laptop Solutions\Downloads\NAVTTC-A3>
```